

Two-Level Electronic Systems

Companion notes to Computational Set 2

Jay Foley
Department of Chemistry, University of North Carolina Charlotte

CHEM 5200 · Computational Chemistry

Abstract

In Computational Set 1 we developed the formalism of spin 1/2: the Pauli matrices, their eigenvectors, and their commutation relations. Nothing in that development required the system to be a spin. These notes make the general statement precise — *every* quantum system with two relevant states is governed by the same algebra — and then apply it to four concrete chemical problems: the two-orbital LCAO dimer, a spin or chromophore driven by an oscillating field, donor–acceptor electron transfer along a nuclear coordinate, and the ammonia inversion doublet in a static electric field. In each case a $\hat{\sigma}_z$ term measures the energy asymmetry between the two basis states and a $\hat{\sigma}_x$ (or $\hat{\sigma}_y$) term measures the coupling between them. We close by asking where a genuine $\hat{\sigma}_y$ contribution can come from, and by identifying the two models — Jaynes–Cummings and Holstein — that these examples become once a boson mode is added.

How to use these notes

These notes are the analytical companion to the Jupyter notebook `notebooks/two_level_systems.ipynb`. The notebook builds the same results numerically and asks you to write the code; the notes carry the derivations that are only sketched there. Read [Sections 1](#) and [2](#) before starting the notebook, and use the appendices whenever a step in the main text is quoted rather than shown. The problems in [Section 9](#) are the same ones that close the notebook.

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1 Every two-level Hamiltonian is a spin

Chemistry supplies two-level systems in abundance. A bond in a minimal basis has two atomic orbitals; a proton in an NMR magnet has two spin projections; a donor–acceptor complex has the electron on one site or the other; ammonia has its umbrella up or down; a chromophore in a weak laser field has a ground and a first excited state. In each case the relevant Hilbert space is two dimensional, and in each case — as we now show — the Hamiltonian is a linear combination of the identity and the three Pauli matrices.

1.1 The Pauli matrices are a complete basis

Let $|\alpha\rangle$ and $|\beta\rangle$ be an orthonormal pair spanning the space. Which two physical states these are will change from section to section; the algebra will not. Any Hamiltonian is a 2×2 Hermitian matrix,

$$\hat{H} = \begin{bmatrix} H_{\alpha\alpha} & H_{\alpha\beta} \\ H_{\alpha\beta}^* & H_{\beta\beta} \end{bmatrix}, \quad H_{\alpha\alpha}, H_{\beta\beta} \in \mathbb{R}, \quad (1)$$

which carries four real degrees of freedom: the two diagonal entries and the real and imaginary parts of $H_{\alpha\beta}$. The set $\{\hat{1}, \hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$ carries four real degrees of freedom as well, and it is straightforward to check that these four matrices are linearly independent. They therefore span the space of Hermitian 2×2 matrices, and we may always write

The Pauli decomposition

$$\hat{H} = \varepsilon_0 \hat{1} + \frac{1}{2} (d_x \hat{\sigma}_x + d_y \hat{\sigma}_y + d_z \hat{\sigma}_z) \equiv \varepsilon_0 \hat{1} + \frac{1}{2} \mathbf{d} \cdot \hat{\sigma}, \quad (2)$$

with ε_0 and the three components of $\mathbf{d}$ all real.

Writing Eq. (2) out in components,

$$\hat{H} = \begin{bmatrix} \varepsilon_0 + \frac{d_z}{2} & \frac{d_x - id_y}{2} \\ \frac{d_x + id_y}{2} & \varepsilon_0 - \frac{d_z}{2} \end{bmatrix}, \quad (3)$$

so that each piece can be read off and given a name.

Component	Definition	Physical meaning
ε_0	$\frac{1}{2} (H_{\alpha\alpha} + H_{\beta\beta})$	mean energy; shifts everything, changes nothing
d_z	$H_{\alpha\alpha} - H_{\beta\beta}$	asymmetry : the energy gap between the basis states
d_x	$2 \operatorname{Re} H_{\alpha\beta}$	coupling : the real part of the mixing element
d_y	$-2 \operatorname{Im} H_{\alpha\beta}$	coupling : the imaginary part

That table is the thesis of these notes. Everything that follows is an exercise in identifying, for some piece of chemistry, what plays the role of the asymmetry and what plays the role of the coupling.

1.2 Extracting the components

The Pauli matrices satisfy the two trace identities

$$\operatorname{Tr}(\hat{\sigma}_x) = \operatorname{Tr}(\hat{\sigma}_y) = \operatorname{Tr}(\hat{\sigma}_z) = 0, \quad \operatorname{Tr}(\hat{\sigma}_i \hat{\sigma}_j) = 2 \delta_{ij}, \quad (4)$$

which follow immediately from the product rule $\hat{\sigma}_i \hat{\sigma}_j = \delta_{ij} \hat{1} + i \epsilon_{ijk} \hat{\sigma}_k$ (derived in [Appendix A](#)). Multiplying [Eq. \(2\)](#) by $\hat{\sigma}_i$ and tracing therefore isolates a single component:

$$\varepsilon_0 = \frac{1}{2} \text{Tr}(\hat{H}), \quad d_i = \text{Tr}(\hat{\sigma}_i \hat{H}). \quad (5)$$

In the notebook

[Equation \(5\)](#) is the function `pauli_decomposition(H)`, and [Eq. \(2\)](#) is `build_two_level(e0, dz, dx, dy)`. The notebook verifies that the two are exact inverses of one another on 500 randomly generated Hermitian matrices — a much stronger check than any single worked example, because it tests completeness of the basis rather than one lucky case.

1.3 A first example: the spin operators themselves

As a sanity check, take the operators from Computational Set 1. Since $\hat{S}_z = \frac{\hbar}{2} \hat{\sigma}_z$, applying [Eq. \(5\)](#) gives $\varepsilon_0 = 0$ and $\mathbf{d} = (0, 0, \hbar)$. Similarly $\hat{S}_x$ gives $\mathbf{d} = (\hbar, 0, 0)$ and $\hat{S}_y$ gives $\mathbf{d} = (0, \hbar, 0)$. The three spin operators are simply the three coordinate axes of the $\mathbf{d}$ space, which is why that space is worth taking seriously as a geometric object — the subject of the next section.

2 Solving the general two-level problem

2.1 Eigenvalues without diagonalizing

The decomposition of [Section 1](#) pays for itself immediately. Using the product rule for Pauli matrices one finds ([Appendix A](#))

$$(\mathbf{d} \cdot \hat{\sigma})^2 = |\mathbf{d}|^2 \hat{1}. \quad (6)$$

A matrix whose square is a multiple of the identity can only have eigenvalues $\pm |\mathbf{d}|$, so the spectrum of $\hat{H}$ follows with no linear algebra at all:

Splitting

$$E_{\pm} = \varepsilon_0 \pm \frac{|\mathbf{d}|}{2} = \varepsilon_0 \pm \frac{1}{2} \sqrt{d_x^2 + d_y^2 + d_z^2}, \quad E_+ - E_- = |\mathbf{d}|. \quad (7)$$

This one formula is the bonding–antibonding splitting, the generalized Rabi frequency, the adiabatic gap at an avoided crossing, the Davydov splitting of an excitonic dimer, and the inversion doublet of ammonia. It is worth pausing on what it says. Writing $d_{\perp} \equiv \sqrt{d_x^2 + d_y^2}$ for the coupling magnitude,

$$E_+ - E_- = \sqrt{d_z^2 + d_{\perp}^2} \geq d_{\perp}. \quad (8)$$

The asymmetry and the coupling add *in quadrature*, so the splitting can never fall below the coupling. Even when the two basis states are exactly degenerate, they are split by $d_{\perp}$. This is the **non-crossing rule**: two states that are coupled cannot become degenerate by tuning one parameter.

2.2 The mixing angle

Write $\mathbf{d}$ in spherical polar coordinates,

$$d_z = |\mathbf{d}| \cos \theta, \quad d_x = |\mathbf{d}| \sin \theta \cos \varphi, \quad d_y = |\mathbf{d}| \sin \theta \sin \varphi, \quad (9)$$

so that

$$\tan \theta = \frac{d_{\perp}}{d_z}, \quad \varphi = \text{atan2}(d_y, d_x). \quad (10)$$

Substituting into Eq. (3) and solving (Appendix B) gives the eigenvectors:

Eigenvectors	
$ +\rangle = \cos \frac{\theta}{2} \alpha\rangle + e^{i\varphi} \sin \frac{\theta}{2} \beta\rangle,$	(11)
$ -\rangle = -e^{-i\varphi} \sin \frac{\theta}{2} \alpha\rangle + \cos \frac{\theta}{2} \beta\rangle.$	(12)

The angle θ is the single most useful number in two-level physics: it measures how thoroughly the coupling has scrambled the basis states. The population of $|\alpha\rangle$ in the upper state is $\cos^2(\theta/2)$ and in the lower state $\sin^2(\theta/2)$.

Regime	θ	Eigenstates are...
$d_z \gg d_{\perp} > 0$	$\rightarrow 0$	essentially $ \alpha\rangle, \beta\rangle$; coupling is a perturbation
$d_z = 0$	$= \pi/2$	exactly 50/50; maximally delocalized
$-d_z \gg d_{\perp} > 0$	$\rightarrow \pi$	localized again, with the labels exchanged

Note the half-angles in Eq. (12). As d_z sweeps from $+\infty$ to $-\infty$ at fixed $d_{\perp}$, θ runs from 0 to π and $|+\rangle$ evolves continuously from $|\alpha\rangle$ to $|\beta\rangle$. That continuous exchange of character, without the levels ever touching, is an avoided crossing. Figure 1 shows both halves of the story.

2.3 Dynamics: the Bloch equation

For a normalized state $|\psi\rangle$ define the **Bloch vector**

$$\mathbf{s} = \langle \hat{\sigma} \rangle = \left(\langle \psi | \hat{\sigma}_x | \psi \rangle, \langle \psi | \hat{\sigma}_y | \psi \rangle, \langle \psi | \hat{\sigma}_z | \psi \rangle \right). \quad (13)$$

Three real numbers, and for any pure state $|\mathbf{s}| = 1$ exactly, so the state lives on the surface of a unit sphere. The north pole is $|\alpha\rangle$, the south pole is $|\beta\rangle$, the equator holds the equally weighted superpositions, and longitude on the equator is the relative phase. Comparing Eq. (13) with Eq. (12), the eigenstate $|+\rangle$ sits at polar angles (θ, φ) — that is, $\mathbf{s}_+ = \hat{\mathbf{d}}$. **The eigenstates are the two points where the state vector is parallel or antiparallel to $\mathbf{d}$.**

Applying the Heisenberg equation to each Pauli matrix and using $[\hat{\sigma}_a, \hat{\sigma}_b] = 2i\epsilon_{abc}\hat{\sigma}_c$ gives a strikingly simple result, worked through in full in Appendix C:

Bloch equation
$\frac{d\mathbf{s}}{dt} = \frac{1}{\hbar} \mathbf{d} \times \mathbf{s}.$

The Bloch vector *precesses about* $\mathbf{d}$ at angular frequency $|\mathbf{d}|/\hbar$. Three consequences deserve to be stated explicitly.

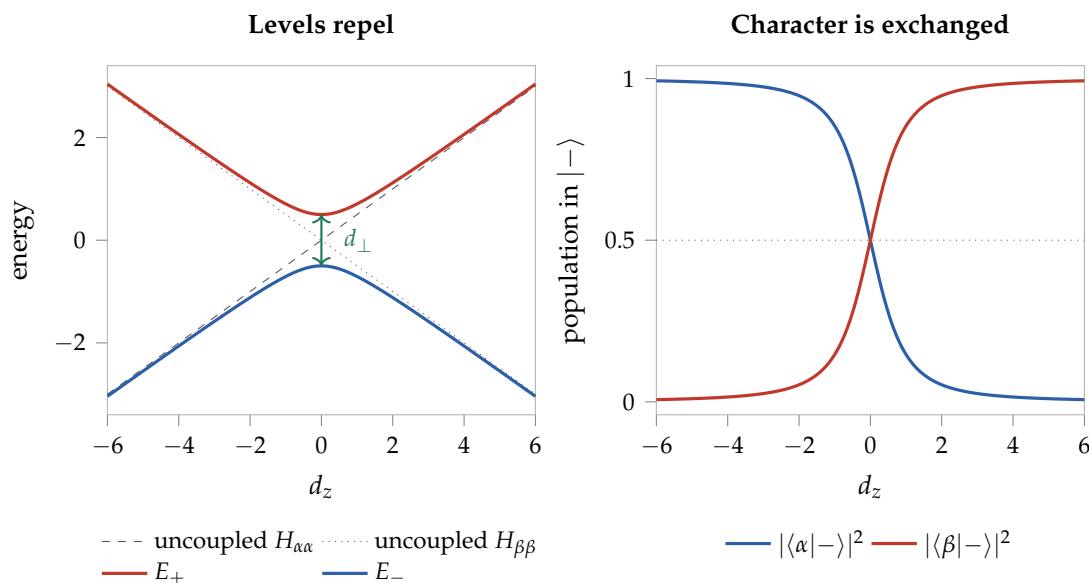


Figure 1: The two faces of an avoided crossing, for fixed coupling $d_\perp = 1$. **Left:** the uncoupled diagonal energies (grey) cross at $d_z = 0$; the eigenvalues of Eq. (7) do not, and are separated there by exactly $d_\perp$. **Right:** the composition of the lower eigenstate, from Eq. (12). The exchange of character happens over a window of width $\sim d_\perp$ in d_z — a stronger coupling makes the switch *more* gradual, not less.

1. **Energy is conserved.** A cross product is perpendicular to both arguments, so $\mathbf{d} \cdot \mathbf{s}$ is constant — which is exactly the statement that $\langle \hat{H} \rangle = \varepsilon_0 + \frac{1}{2} \mathbf{d} \cdot \mathbf{s}$ does not change.
2. **Eigenstates do not move.** If $\mathbf{s} \parallel \mathbf{d}$ the cross product vanishes. Stationary states are stationary.
3. **The precession frequency is the Bohr frequency.** Since $|\mathbf{d}| = E_+ - E_-$, the rate is $(E_+ - E_-)/\hbar$.

Notice that ε_0 is absent from Eq. (14). An overall energy shift produces an overall phase, which no measurement on this system can detect.

In the notebook

Equation (13) is `bloch_vector(psi)`; time evolution is `propagate(H, psi0, times)`, which phases the eigenbasis coefficients rather than integrating an ODE. The notebook checks Eq. (14) indirectly, by confirming that the projection $\mathbf{s} \cdot \mathbf{d}$ is constant to machine precision along a trajectory.

3 Example I: the two-orbital LCAO dimer

The simplest molecular orbital problem in chemistry is two atoms carrying one atomic orbital each. Identify $|A\rangle \equiv |\alpha\rangle$ and $|B\rangle \equiv |\beta\rangle$.

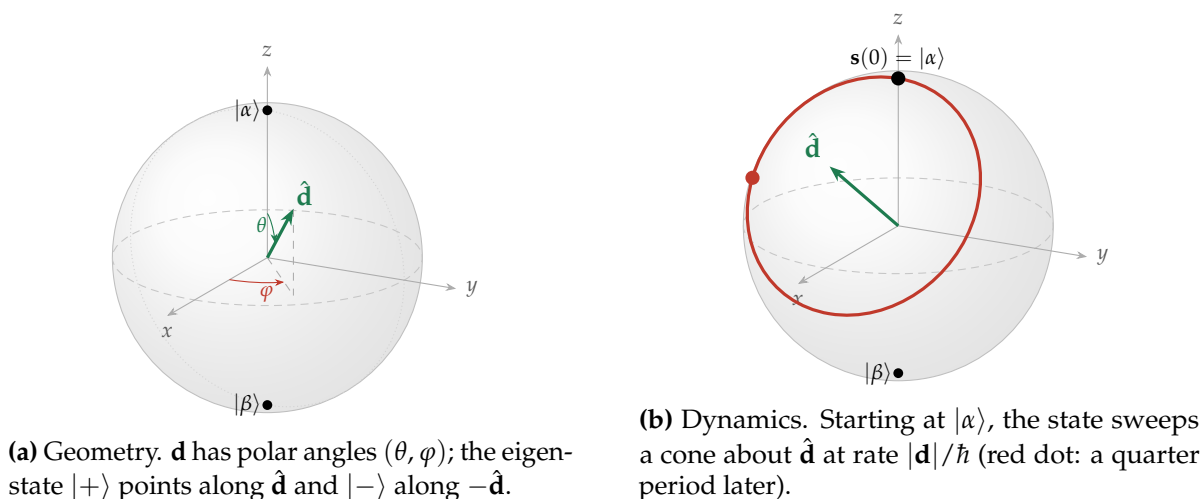


Figure 2: The Bloch sphere. Because the trajectory is a cone about $\hat{\mathbf{d}}$, a state prepared at the north pole reaches the south pole only if $\hat{\mathbf{d}}$ lies in the equatorial plane — which is precisely the resonance condition of [Section 4](#).

A word on orthogonality

Atomic orbitals on neighbouring atoms overlap, $\langle A|B\rangle = S \neq 0$, which strictly gives a generalized eigenvalue problem $\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}$. Throughout this section we assume the basis has already been symmetrically (Löwdin) orthogonalized, $\mathbf{S}^{-1/2}\mathbf{H}\mathbf{S}^{-1/2}$, so that $\mathbf{S} = \mathbf{1}$. This is exactly the assumption Hückel theory makes, and it is what lets us use the machinery of [Section 2](#) unmodified. The general overlap case is taken up in the Basis Sets set.

3.1 Identifying $\mathbf{d}$

In the orthogonalized basis the matrix elements carry their traditional Hückel names,

$$\hat{H} = \begin{bmatrix} \alpha_A & \beta \\ \beta & \alpha_B \end{bmatrix}, \quad (15)$$

where $\alpha_X = \langle X|\hat{H}|X\rangle$ is the *Coulomb integral* — the energy of an electron localized in the orbital on atom X , more negative for a more electronegative atom — and $\beta = \langle A|\hat{H}|B\rangle$ is the *resonance* or *hopping integral*, negative by convention and falling off roughly exponentially with internuclear separation. Applying [Eq. \(5\)](#),

LCAO dimer

$$\epsilon_0 = \frac{\alpha_A + \alpha_B}{2}, \quad d_z = \underbrace{\alpha_A - \alpha_B}_{\text{electronegativity difference}}, \quad d_x = \underbrace{2\beta}_{\text{twice the hopping}}, \quad d_y = 0. \quad (16)$$

The absence of d_y is not an accident of this example; it is a consequence of time-reversal symmetry, taken up in [Section 7](#).

From Eq. (7) the molecular orbital energies are

$$E_{\pm} = \frac{\alpha_A + \alpha_B}{2} \pm \sqrt{\left(\frac{\alpha_A - \alpha_B}{2}\right)^2 + \beta^2}, \quad (17)$$

and the mixing angle is $\tan \theta = 2\beta/(\alpha_A - \alpha_B)$.

3.2 The two limits

Homonuclear: $d_z = 0$. For H_2^+ , or for the π system of ethylene, $\alpha_A = \alpha_B \equiv \alpha$ and the Hamiltonian is *pure* $\hat{\sigma}_x$. Then $\theta = \pi/2$, the eigenvectors are $\frac{1}{\sqrt{2}}(|A\rangle \pm |B\rangle)$, and $E_{\pm} = \alpha \pm |\beta|$: a bonding orbital at $\alpha + \beta$ (recall $\beta < 0$) and an antibonding orbital at $\alpha - \beta$, split by $2|\beta|$.

It is worth noticing that this Hamiltonian is $\hat{S}_x$ up to a scale and a shift. The bonding and antibonding MOs of a homonuclear diatomic are, as vectors, precisely the eigenvectors of $\hat{S}_x$ you computed in Computational Set 1.

Strongly heteronuclear: $|d_z| \gg |d_x|$. Now $\theta \rightarrow 0$, the orbitals barely mix, and the doubly occupied lower MO sits almost entirely on the more electronegative atom. This is the ionic limit. Everything in between is a polar covalent bond, and the single dimensionless parameter controlling where you sit is

$$\frac{|d_x|}{|d_z|} = \frac{2|\beta|}{|\alpha_A - \alpha_B|}. \quad (18)$$

Mixed-valence chemistry

Ratio (18) is the physical content of the **Robin–Day** classification of mixed-valence complexes [3]. Class I ($|\beta| \ll |d_z|/2$) means fully trapped valences and no intervalence band; Class III ($|\beta| \gg |d_z|/2$) means a completely delocalized, symmetric ion — the Creutz–Taube complex $[(\text{NH}_3)_5\text{Ru-pyrazine-Ru}(\text{NH}_3)_5]^{5+}$ being the canonical case [4]. Class II is the interesting middle, where the intervalence charge-transfer band appears and Marcus theory (Section 5) governs the thermal transfer rate.

3.3 A worked case

Take $\alpha_A = -11.4$ eV, $\alpha_B = -13.6$ eV (atom B the more electronegative), and $\beta = -2.7$ eV. Then $\varepsilon_0 = -12.5$ eV, $d_z = 2.2$ eV, and $d_x = -5.4$ eV, so

$$d_{\perp} = |2\beta| = 5.4 \text{ eV}, \quad |\mathbf{d}| = \sqrt{2.2^2 + 5.4^2} \text{ eV} = 5.831 \text{ eV}, \quad (19)$$

and, since $d_x < 0$ puts the transverse component along $-\hat{x}$,

$$\theta = \text{atan2}(d_{\perp}, d_z) = \text{atan2}(5.4, 2.2) = 67.83^\circ, \quad \varphi = \pi. \quad (20)$$

The MOs come out at -15.415 eV (bonding) and -9.585 eV (antibonding). The bonding MO carries $\sin^2(\theta/2) = 0.311$ of its density on A and 0.689 on B — polarized toward the electronegative atom, as chemical intuition demands, but nowhere near the ionic limit because $2|\beta|$ is more than twice $|d_z|$.

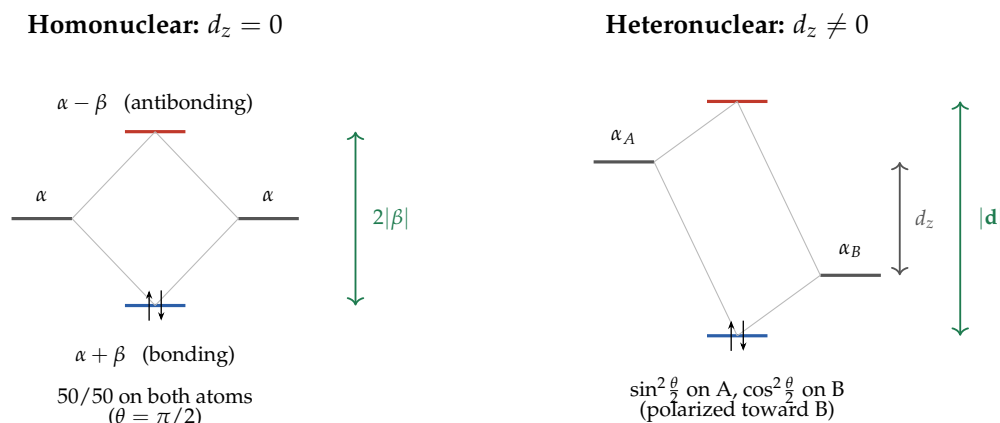


Figure 3: Molecular orbital correlation diagrams for the two-orbital dimer (recall $\beta < 0$, so $\alpha + \beta$ is the lower level). The splitting is $|\mathbf{d}| = \sqrt{d_z^2 + 4\beta^2}$ in both cases; what changes is how the lower MO distributes itself between the atoms, which is governed entirely by the mixing angle θ of Eq. (10).

Figure 1 is exactly the picture of this system as d_z is scanned at fixed β : a bond that is ionic at one extreme, purely covalent at $d_z = 0$, and ionic with the opposite polarity at the other extreme.

In the notebook

Part 3 of the notebook builds this Hamiltonian, decomposes it, reproduces the numbers above, and then scans d_z to generate Fig. 1. One of the test cells asserts that the gap never falls below $2|\beta|$ across the entire scan — a numerical statement of the non-crossing rule, Eq. (8).

4 Example II: a driven two-level system

4.1 The lab-frame Hamiltonian

A spin-1/2 nucleus in a static field $B_0 \hat{z}$ has the Zeeman Hamiltonian

$$\hat{H}_0 = -\gamma B_0 \hat{S}_z = \frac{\hbar \omega_0}{2} \hat{\sigma}_z, \quad \omega_0 \equiv -\gamma B_0, \quad (21)$$

pure $\hat{\sigma}_z$, with the two spin states split by the Larmor energy $\hbar \omega_0$. For a proton at 11.7 T this is 500 MHz, the number printed on the front of the spectrometer. Nothing happens under $\hat{H}_0$ alone: $|\alpha\rangle$ and $|\beta\rangle$ are eigenstates and the Bloch vector merely spins about $\hat{z}$.

To make something happen, add a weak field rotating in the xy plane at frequency ω — the RF coil in NMR, or, for an electronic transition in the dipole approximation, the optical field of a laser:

$$\hat{H}(t) = \frac{\hbar \omega_0}{2} \hat{\sigma}_z + \frac{\hbar \omega_1}{2} [\cos(\omega t + \varphi) \hat{\sigma}_x + \sin(\omega t + \varphi) \hat{\sigma}_y], \quad \omega_1 \equiv -\gamma B_1. \quad (22)$$

This is genuinely time dependent, so the machinery of [Section 2](#) does not apply directly. One transformation fixes that.

4.2 The rotating frame

Move to a frame rotating about z at the drive frequency by defining $|\tilde{\psi}\rangle = R(t)|\psi\rangle$ with $R(t) = e^{+i\omega t \hat{\sigma}_z/2}$. Substituting into the Schrödinger equation (worked out step by step in [Appendix D](#)) gives an effective Hamiltonian $\tilde{H} = R\hat{H}R^\dagger - \frac{\hbar\omega}{2}\hat{\sigma}_z$ which turns out to be *time independent*:

Rotating-frame Hamiltonian

$$\hat{H}_{\text{rot}} = \frac{\hbar}{2} \left[\underbrace{(\omega_0 - \omega)}_{\Delta} \hat{\sigma}_z + \omega_1 (\cos \varphi \hat{\sigma}_x + \sin \varphi \hat{\sigma}_y) \right], \quad (23)$$

that is, $d_z = \hbar\Delta$, $d_x = \hbar\omega_1 \cos \varphi$, $d_y = \hbar\omega_1 \sin \varphi$.

Three features of [Eq. \(23\)](#) are each a piece of real spectroscopic practice.

- **The detuning is the $\hat{\sigma}_z$ knob.** On resonance ($\omega = \omega_0$) it vanishes and $\mathbf{d}$ lies flat in the equatorial plane. Off resonance, $\mathbf{d}$ tilts toward the poles — and by [Fig. 2b](#), a tilted $\mathbf{d}$ means the state can no longer reach the far pole.
- **The pulse phase is the $\hat{\sigma}_y$ knob.** Setting $\varphi = 0$ gives a pure $\hat{\sigma}_x$ term, an “ x -pulse”; $\varphi = \pi/2$ gives a pure $\hat{\sigma}_y$ term, a “ y -pulse”. This is not jargon: φ is literally the phase of the RF pulse, and it is why every pulse-sequence diagram in an NMR textbook labels its pulses x , y , $-x$, $-y$.
- **The tilted-cone geometry of [Section 2.3](#) is now physical.** The polar angle of $\mathbf{d}$ is set by how far off resonance you are, and its azimuth by the pulse phase.

4.3 Rabi oscillations

Apply [Eq. \(7\)](#). The splitting in the rotating frame defines the **generalized Rabi frequency**

$$\Omega = \frac{|\mathbf{d}|}{\hbar} = \sqrt{\Delta^2 + \omega_1^2}, \quad (24)$$

and integrating [Eq. \(14\)](#) from $\mathbf{s}(0) = \hat{z}$ (or equivalently applying [Eq. \(12\)](#) twice) gives

Rabi formula

$$P_\beta(t) = \frac{\omega_1^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right). \quad (25)$$

[Equation \(25\)](#) repays careful reading. The *rate* of oscillation, $\Omega = \sqrt{\Delta^2 + \omega_1^2}$, always *increases* with detuning. But the *amplitude*,

$$P_\beta^{\text{max}} = \frac{\omega_1^2}{\Delta^2 + \omega_1^2}, \quad (26)$$

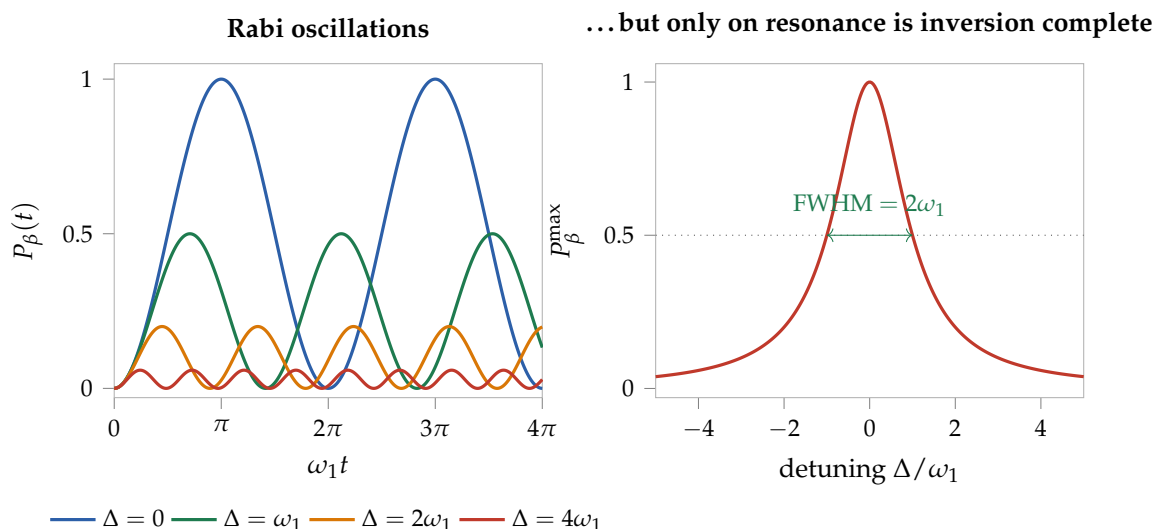


Figure 4: Left: Eq. (25) at four detunings. Detuning makes the oscillation *faster* and *shallower* at the same time. Right: the maximum transferable population, Eq. (26). A pulse of Rabi strength ω_1 can only fully invert spins within roughly $\pm\omega_1$ of resonance — which is what “selective pulse” means.

is a Lorentzian in Δ of half-width ω_1 : complete population inversion is possible only exactly on resonance. Equation (26) is simultaneously the origin of the spectroscopic lineshape and of *pulse selectivity* — a weak, long pulse addresses a narrow band of frequencies; a strong, short pulse excites everything in sight. Figure 4 shows both effects.

4.4 π and $\pi/2$ pulses

On resonance $\mathbf{d}$ lies in the equatorial plane, the precession cone opens into a great circle through both poles, and the state sweeps from $|\alpha\rangle$ to $|\beta\rangle$ and back. Two durations matter:

$$t_\pi = \frac{\pi}{\omega_1} : |\alpha\rangle \rightarrow |\beta\rangle \text{ (inversion)}, \quad t_{\pi/2} = \frac{\pi}{2\omega_1} : |\alpha\rangle \rightarrow \text{the equator (coherence)}. \quad (27)$$

The $\pi/2$ pulse is the one that generates an NMR signal: it converts population difference into transverse coherence, which is what the detection coil sees as the free induction decay. And the pulse phase decides *where* on the equator you land. Using Eq. (14) with $\mathbf{d} = \hbar\omega_1\hat{x}$,

$$x\text{-pulse: } \hat{z} \rightarrow -\hat{y}, \quad y\text{-pulse: } \hat{z} \rightarrow +\hat{x}, \quad (28)$$

two states 90° apart on the equator. Both are “the same pulse” in every respect except phase.

Where this is going: the Jaynes–Cummings model

Write the transverse part of Eq. (23) using the ladder operators $\hat{\sigma}_\pm = \frac{1}{2}(\hat{\sigma}_x \pm i\hat{\sigma}_y)$:

$$\frac{\hbar\omega_1}{2} (\cos \varphi \hat{\sigma}_x + \sin \varphi \hat{\sigma}_y) = \frac{\hbar\omega_1}{2} (e^{-i\varphi} \hat{\sigma}_+ + e^{+i\varphi} \hat{\sigma}_-). \quad (29)$$

The drive *raises* the two-level system while supplying a phase, or *lowers* it while absorbing one. Here the field is a classical number: $\omega_1 \propto B_1$, and it is inexhaustible — the system can cycle forever and the field never notices.

Now promote the field to a quantum degree of freedom. The phase factors become photon annihilation and creation operators, $e^{-i\varphi} \rightarrow \hat{a}$ and $e^{+i\varphi} \rightarrow \hat{a}^\dagger$, and

$$\hat{H}_{\text{JC}} = \frac{\hbar\omega_0}{2}\hat{\sigma}_z + \hbar\omega\hat{a}^\dagger\hat{a} + \hbar g(\hat{a}\hat{\sigma}_+ + \hat{a}^\dagger\hat{\sigma}_-), \quad (30)$$

the **Jaynes–Cummings** Hamiltonian [14] — the single-emitter case of the Tavis–Cummings model built in Computational Set 4. The Hilbert space is no longer two dimensional; it is spin $\otimes$ boson. The classical drive strength ω_1 is replaced by $2g\sqrt{n+1}$, so that even the vacuum ($n = 0$) drives Rabi oscillations. That is vacuum Rabi splitting, and it is the foundation of polaritonic chemistry. Everything in this section is its classical-field limit.

In the notebook

Part 5 verifies Eq. (25) against numerical propagation to 10^{-9} for several detunings, confirms that a resonant π pulse inverts the population exactly, and shows that the x - and y -pulses leave the Bloch vector 90° apart.

5 Example III: donor–acceptor electron transfer

5.1 Diabatic states and a nuclear coordinate

Consider an electron transferring from a donor to an acceptor, $D^-A \rightarrow D-A^-$. The two basis states are the **diabatic** (charge-localized) states $|D\rangle \equiv |\alpha\rangle$ and $|A\rangle \equiv |\beta\rangle$.

What is new here is that their energies are not fixed numbers. Solvent molecules reorganize around whichever site currently holds the charge, so the diabatic energies depend on where the nuclei are. Introduce a single collective nuclear coordinate q — the solvent or reaction coordinate — and model each diabatic surface as a harmonic well of the same force constant k , displaced by $\pm q_0$:

$$V_D(q) = \frac{1}{2}k(q + q_0)^2, \quad V_A(q) = \frac{1}{2}k(q - q_0)^2 + \Delta G^\circ. \quad (31)$$

The donor state is happiest at $q = -q_0$, the acceptor state at $q = +q_0$, and ΔG° is the thermodynamic driving force. The electronic coupling $V_{DA} = \langle D|\hat{H}|A\rangle$ between them is essentially independent of q .

5.2 Identifying d

With $\hat{H}(q)$ built from Eq. (31),

$$d_z(q) = V_D(q) - V_A(q) = 2kq_0q - \Delta G^\circ, \quad d_x = 2V_{DA}, \quad d_y = 0. \quad (32)$$

Defining the **reorganization energy** $\lambda \equiv 2kq_0^2$ — the energy cost of distorting the nuclei from the donor’s optimal geometry to the acceptor’s *without* moving the electron:

Electron transfer

$$d_z(q) = \frac{\lambda}{q_0} q - \Delta G^\circ, \quad d_x = 2V_{DA}. \quad (33)$$

This is the structural heart of the example, and it is worth stating plainly: **the $\hat{\sigma}_z$ coefficient is linear in a nuclear coordinate, and the $\hat{\sigma}_x$ coefficient is a constant.** Compare Eq. (16), where d_z was a fixed electronegativity difference. Here the molecule sweeps its own d_z back and forth as it vibrates, so the parameter scan of Fig. 1 is not a hypothetical — it is something the molecule physically does, many times per picosecond.

5.3 Adiabatic surfaces and the Marcus barrier

The eigenvalues of $\hat{H}(q)$ are the **adiabatic** potential energy surfaces — the Born–Oppenheimer curves the nuclei actually move on:

$$E_{\pm}(q) = \bar{V}(q) \pm \frac{1}{2} \sqrt{d_z(q)^2 + 4V_{DA}^2}, \quad \bar{V}(q) = \frac{1}{2}[V_D(q) + V_A(q)]. \quad (34)$$

Three consequences follow at once from Section 2:

- the diabats cross where $d_z(q^\ddagger) = 0$, that is at $q^\ddagger = \Delta G^\circ q_0 / \lambda$;
- the adiabatic gap there is exactly $|d_x| = 2V_{DA}$;
- the activation energy, measured on the *diabatic* surface from the donor minimum up to the crossing, is

$$\Delta G^\ddagger = V_D(q^\ddagger) - V_D(-q_0) = \frac{(\lambda + \Delta G^\circ)^2}{4\lambda}, \quad (35)$$

the **Marcus** activation free energy [8, 9] (algebra in Appendix E).

Equation (35) is a parabola in the driving force, minimized — and equal to zero — at $\Delta G^\circ = -\lambda$. Since the nonadiabatic rate goes as $k_{ET} \propto |V_{DA}|^2 \exp(-\Delta G^\ddagger/k_B T)$, making a reaction *more* exergonic speeds it up only until $-\Delta G^\circ = \lambda$, and slows it down thereafter. This is the celebrated **inverted region**, whose experimental confirmation was central to the 1992 Nobel Prize in Chemistry. Figures 5 and 6 show the surfaces and the rate.

5.4 Landau–Zener: sweeping through the crossing

Everything above is static — surfaces evaluated at fixed q . Now let the nuclei actually move through the crossing. If the coordinate sweeps linearly in time, $q = vt$, then by Eq. (33)

$$d_z(t) = \alpha t, \quad \alpha = \frac{\lambda v}{q_0} = \left| \frac{d}{dt} (V_D - V_A) \right|, \quad (36)$$

the rate at which the diabatic gap is swept. Starting far to one side in a single diabatic state, two outcomes compete:

- **adiabatic passage** — the system follows the lower adiabatic surface and therefore *changes* diabatic character: the electron transfers;

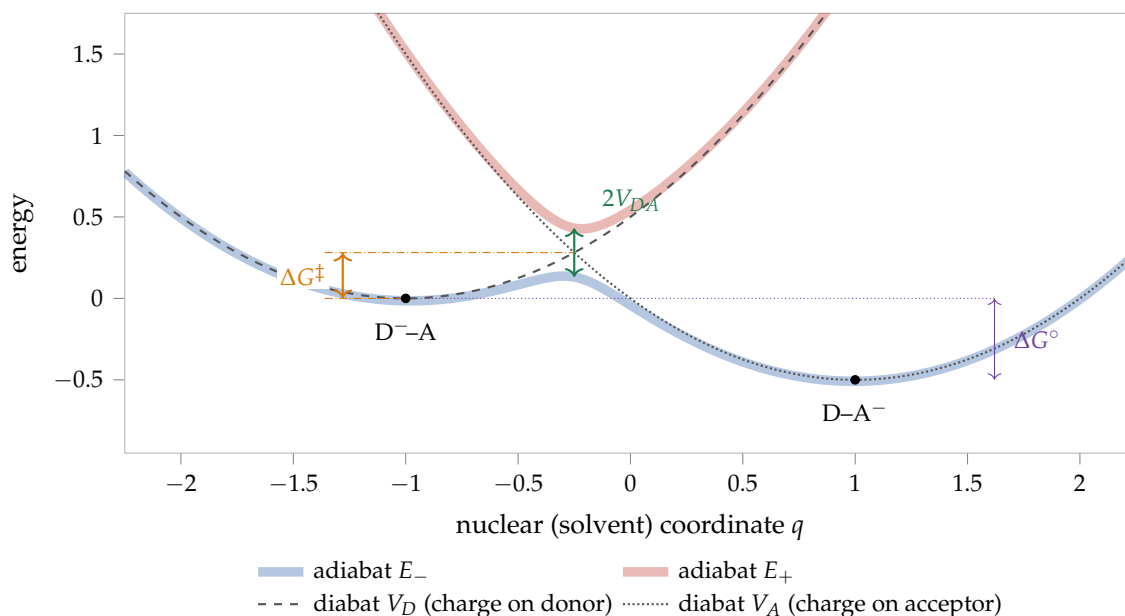


Figure 5: Diabatic and adiabatic surfaces for $k = 1$, $q_0 = 1$ (so $\lambda = 2$), $\Delta G^\circ = -0.5$, $V_{DA} = 0.15$. Away from the crossing the adiabats and diabats coincide — the electron stays put. Near $q^\ddagger = \Delta G^\circ q_0 / \lambda = -0.25$ the coupling splits them by $2V_{DA}$ and the character of each adiabatic surface switches over.

- **diabatic passage** — the system blasts straight through the avoided crossing, staying on the same diabats: no transfer.

Landau and Zener solved this exactly in 1932 [11, 10]:

Landau–Zener	
$P_{\text{diabatic}} = \exp\left(-\frac{2\pi V_{DA}^2}{\hbar \alpha}\right).$	(37)

Slow sweeps or strong coupling give adiabatic passage ($P \rightarrow 0$); fast sweeps or weak coupling give diabatic passage ($P \rightarrow 1$). Note the combination in the exponent: (coupling)² divided by (sweep rate). Expanding for small V_{DA} ,

$$1 - P_{\text{diabatic}} \approx \frac{2\pi V_{DA}^2}{\hbar \alpha}, \quad (38)$$

which is the same $|V_{DA}|^2$ proportionality that appears in Fermi's golden rule and in the nonadiabatic Marcus rate — the two results are the same physics seen from two directions.

Where this is going: the Holstein model

Collecting Eq. (33),

$$\hat{H}(q) = \bar{V}(q) \hat{\mathbb{1}} + \frac{1}{2} \left[\frac{\lambda}{q_0} q - \Delta G^\circ \right] \hat{\sigma}_z + V_{DA} \hat{\sigma}_x. \quad (39)$$

We treated q as a *classical parameter* — a number to dial, or to sweep linearly in time. But

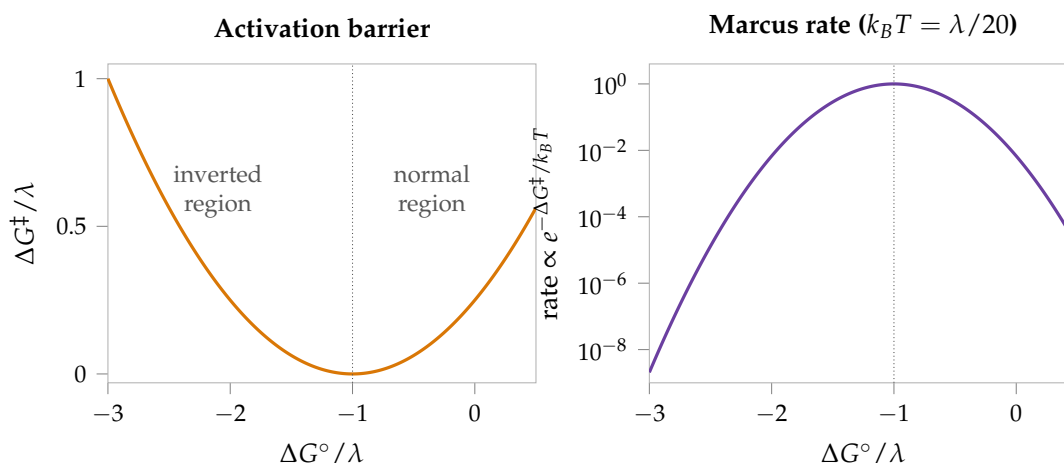


Figure 6: The Marcus inverted region. The barrier (35) is a parabola in the driving force, so the rate is *not* monotonic: past $-\Delta G^\circ = \lambda$ a more exergonic reaction is a slower one.

q is a nuclear coordinate, and nuclei are quantum. Promote it to a harmonic oscillator, $\hat{q} \propto (\hat{a} + \hat{a}^\dagger)$, and

$$\hat{H}_{\text{Holstein}} = \frac{\varepsilon}{2} \hat{\sigma}_z + V_{DA} \hat{\sigma}_x + \hbar\omega \hat{a}^\dagger \hat{a} + \hbar\omega g (\hat{a} + \hat{a}^\dagger) \hat{\sigma}_z, \quad (40)$$

the **Holstein**, or spin-boson, Hamiltonian [15, 16].

Set Eq. (40) beside Eq. (30) and notice how different they are despite sharing a two-dimensional electronic space and one boson:

Model	Coupling operator	The boson. . .
Jaynes–Cummings	$\hat{a}\hat{\sigma}_+ + \hat{a}^\dagger\hat{\sigma}_-$	flips the two-level system
Holstein	$(\hat{a} + \hat{a}^\dagger) \hat{\sigma}_z$	shifts its energies

One gives polaritons and vacuum Rabi splitting; the other gives vibronic progressions, Franck–Condon factors, and polaron formation. You have now built the single-spin ancestor of both.

6 Example IV: the ammonia inversion doublet

The three examples so far built their two-level space from two places an electron can be. This one builds it from two *shapes the same molecule can have*.

6.1 Tunneling as an off-diagonal matrix element

Ammonia is pyramidal, and the nitrogen can pass through the plane of the three hydrogens like an umbrella turning inside out. Label the two pyramidal geometries $|L\rangle$ (umbrella down) and $|R\rangle$ (umbrella up). Classically they are separated by a barrier of roughly 2000/cm and the molecule would be stuck in one of them. Quantum mechanically the nitrogen **tunnels**, and

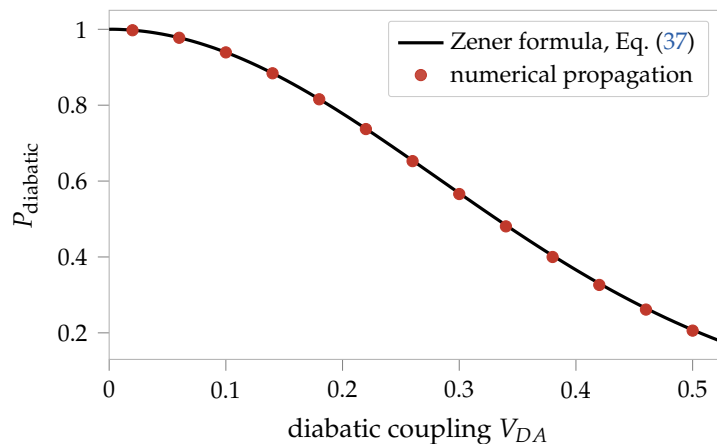


Figure 7: Equation (37) against direct numerical propagation through the crossing ($\hbar = 1$, sweep rate $\alpha = 1$). The numerical points come from the notebook, which slices the sweep into 60000 steps and evaluates $\hat{H}$ at the midpoint of each. The residual disagreement, at most 0.004, is not step error: Zener's result is exact only for a sweep running from $t = -\infty$ to $+\infty$, and any truncated sweep leaves a little interference behind.

tunneling is nothing other than an off-diagonal matrix element:

$$\hat{H} = \begin{bmatrix} 0 & -\Delta_t \\ -\Delta_t & 0 \end{bmatrix} = -\Delta_t \hat{\sigma}_x. \quad (41)$$

Pure $\hat{\sigma}_x$. The eigenstates are $\frac{1}{\sqrt{2}}(|L\rangle \pm |R\rangle)$, split by $2\Delta_t$: the **inversion doublet**, measured at

$$2\Delta_t = 0.7934/\text{cm} = 23.786 \text{ GHz}, \quad (42)$$

a microwave transition — the line Gordon, Zeiger and Townes used to build the first **maser** in 1954, three years before the laser [13].

The conceptual point is worth dwelling on. The energy eigenstates of ammonia are *not* the pyramidal geometries; they are delocalized over both. A molecule prepared as $|L\rangle$ is a superposition of the two eigenstates, so by Section 2.3 it oscillates: the umbrella flips back and forth at 23.786 GHz.

6.2 Adding a $\hat{\sigma}_z$ term: the Stark effect

Switch on a static electric field $\mathcal{E}$ along the molecular axis. The two pyramidal geometries have dipole moments pointing in *opposite* directions, so the field stabilizes one and destabilizes the other:

Ammonia in a field

$$\hat{H} = \mu\mathcal{E} \hat{\sigma}_z - \Delta_t \hat{\sigma}_x \quad \implies \quad d_z = 2\mu\mathcal{E}, \quad d_x = -2\Delta_t, \quad (43)$$

with $\mu = 1.47 \text{ D}$ the NH_3 dipole moment.

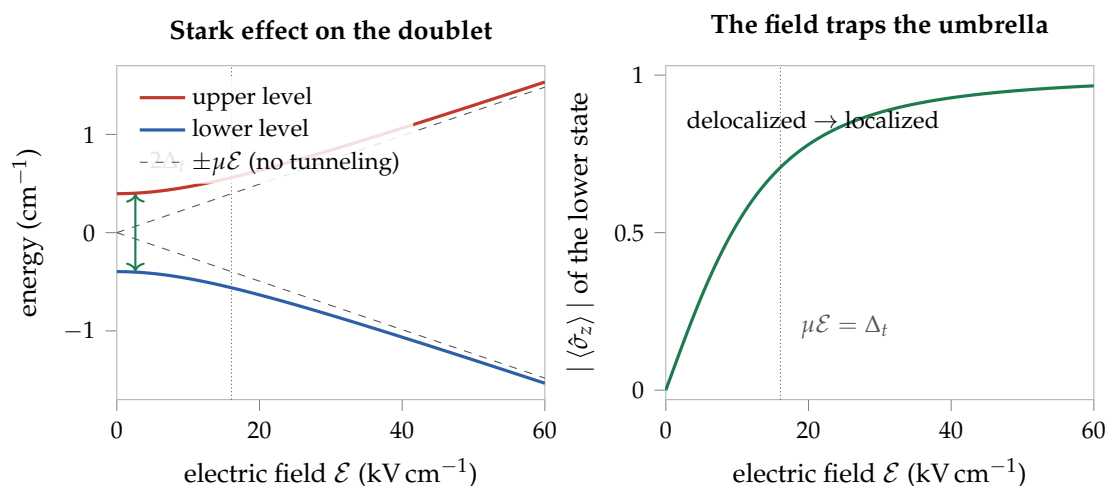


Figure 8: Ammonia in a static electric field, from Eq. (43). **Left:** the levels start split by the tunneling doublet $2\Delta_t = 0.7934/\text{cm}$ and approach the classical Stark asymptotes $\pm\mu\mathcal{E}$ at high field. **Right:** the localization $|\langle \hat{\sigma}_z \rangle| = |d_z|/|\mathbf{d}| = |\cos \theta|$ of the lower state, running from 0 (equal amplitude on both umbrella geometries) to 1 (trapped on one side).

Structurally this is Eq. (16) again — but now *you* control d_z with a knob on the bench. The gap is

$$E_+ - E_- = \sqrt{(2\mu\mathcal{E})^2 + (2\Delta_t)^2}, \quad (44)$$

quadratic in $\mathcal{E}$ at low field (a second-order Stark effect) and crossing over to linear at high field. The crossover occurs where the two terms are comparable, $\mu\mathcal{E} = \Delta_t$, that is at

$$\mathcal{E}_{\text{cross}} = \frac{\Delta_t}{\mu} = 16.07 \text{ kV/cm}, \quad (45)$$

at which point the gap has grown by exactly $\sqrt{2}$. Turn the field well past this and $\theta \rightarrow 0$: the field literally traps the umbrella on one side, converting a delocalized molecule into a localized one. Figure 8 shows both halves.

A puzzle worth posing to yourself

At zero field the ground state is $\frac{1}{\sqrt{2}}(|L\rangle + |R\rangle)$, which has *zero* dipole moment — yet NH_3 is a famously polar molecule with $\mu = 1.47 \text{ D}$. Both statements are true. The resolution turns on what a dipole-moment measurement actually averages over, and on what timescale; see Problem 6 in Section 9.

7 Where does $\hat{\sigma}_y$ come from?

Of the four examples, only the driven system of Section 4 carried a $\hat{\sigma}_y$ term. The LCAO dimer, the electron-transfer complex and ammonia all had $d_y = 0$. That is not a coincidence, and understanding why is worth a short section.

7.1 For one static Hamiltonian, $\hat{\sigma}_y$ is a choice of convention

The overall phase of a basis vector is a convention, not physics. Rephase the second basis state,

$$|\beta\rangle \longrightarrow e^{i\chi} |\beta\rangle, \quad \text{i.e.} \quad \hat{U} = \text{diag}(1, e^{i\chi}), \quad \hat{H}' = \hat{U}^\dagger \hat{H} \hat{U}. \quad (46)$$

This leaves the diagonal untouched and multiplies the off-diagonal element by $e^{i\chi}$. Since $H_{\alpha\beta} = \frac{1}{2}(d_x - id_y) = \frac{1}{2}|d_\perp|e^{-i\varphi}$, choosing $\chi = \varphi$ makes $H'_{\alpha\beta}$ real and positive:

$$d'_x = d_\perp = \sqrt{d_x^2 + d_y^2}, \quad d'_y = 0, \quad d'_z = d_z. \quad (47)$$

The transverse components have simply been rotated into one another. The spectrum, being unitarily invariant, is unchanged — as it must be, since Eq. (7) depends on d_x and d_y only through $d_\perp$.

The gauge statement

For a *single, static* two-level Hamiltonian, a $\hat{\sigma}_y$ term carries no physical content: it can always be rephased away, leaving a real symmetric matrix. Every observable depends on d_x and d_y only through $d_\perp = \sqrt{d_x^2 + d_y^2}$.

7.2 Three ways to make it physical

$\hat{\sigma}_y$ acquires meaning only *relative to something else* that fixes the phase convention. There are three such situations, and each is worth recognizing on sight.

1. A time-dependent drive. In Section 4 the pulse phase φ moved $\mathbf{d}$ around the equatorial plane. Any *single* pulse may be declared an “ x -pulse” by fiat — but once that choice is made, the phase of the *second* pulse in a sequence is physical and measurable. This is exactly what two-dimensional NMR and Ramsey interferometry exploit (Problem 5).

2. Broken time-reversal symmetry. For a spinless electronic Hamiltonian in a basis of real orbitals with no magnetic field, time-reversal symmetry forces $\hat{H}$ to be real symmetric, so $d_y = 0$ automatically — which is why Sections 3 and 5 had none. To get an unremovable $\hat{\sigma}_y$ you must break that symmetry: a magnetic field, which attaches a Peierls phase $\beta \rightarrow \beta e^{i\varphi_B}$ to each hopping integral; circularly polarized light; or **spin-orbit coupling**, which is why the effective 2×2 Hamiltonian of a Kramers doublet in a heavy-atom complex generally carries all three Pauli components.

3. Three or more coupled states. This is the subtle one. With N basis states you have $N - 1$ useful rephasings available (the overall phase does nothing), so you can kill at most $N - 1$ hopping phases. A ring of three sites has three bonds and only two usable phases: one combination survives. What survives is the phase accumulated around the closed loop — a magnetic flux — and it is genuinely gauge invariant. This is the origin of the Aharonov–Bohm effect and, in a different guise, of the Berry phase. Problem 4 works out the dimer and the triangle side by side.

The practical rule

When you meet a real electronic-structure Hamiltonian in a real orbital basis, expect $\hat{\sigma}_z$ and $\hat{\sigma}_x$ only. A $\hat{\sigma}_y$ term is a signal that a field, a spin–orbit interaction, a time-dependent drive, or a closed loop is somewhere in the problem.

8 Summary

Every system in these notes is the same equation, $\hat{H} = \varepsilon_0 \hat{1} + \frac{1}{2} \mathbf{d} \cdot \hat{\sigma}$. Only the dictionary changes.

System	$ \alpha\rangle, \beta\rangle$	d_z (asymmetry)	d_x, d_y (coupling)	$ \mathbf{d} $ is called
LCAO dimer (Section 3)	AO on atom A / atom B	$\alpha_A - \alpha_B$ (electronegativity)	2β (resonance inte- gral)	bonding– antibonding gap
Driven spin (Section 4)	spin up / down	$\hbar(\omega_0 - \omega)$ (detuning)	$\hbar\omega_1 e^{\pm i\varphi}$ (drive $\times$ phase)	$\hbar\Omega$, generalized Rabi
Electron transfer (Section 5)	charge on D / on A	$(\lambda/q_0)q - \Delta G^\circ$ (nuclear coordinate!)	$2V_{DA}$ (electronic cou- pling)	adiabatic gap
NH ₃ inversion (Section 6)	umbrella down / up	$2\mu\mathcal{E}$ (Stark field)	$-2\Delta_t$ (tunneling)	inversion doublet

Table 1: One Hamiltonian, four molecules.

And the physics is always the same three statements:

1. **Splitting.** $E_{\pm} = \varepsilon_0 \pm \frac{1}{2}|\mathbf{d}|$. Because d_z and $d_{\perp}$ add in quadrature, coupled levels never cross; the closest they come is $d_{\perp}$.
2. **Mixing.** $\tan \theta = d_{\perp}/d_z$ decides whether the eigenstates are localized on the basis states or delocalized between them. The switch happens over a window of width $\sim d_{\perp}$.
3. **Dynamics.** $\dot{\mathbf{s}} = \hbar^{-1} \mathbf{d} \times \mathbf{s}$. The state precesses about $\mathbf{d}$ at the Bohr frequency $|\mathbf{d}|/\hbar$.

Faced with a new two-level problem — an exciton dimer, a superconducting qubit, a Kramers doublet, a curve crossing in a photochemical reaction — the first move is always to find $\mathbf{d}$.

What comes next

Both models flagged along the way live in a *larger* Hilbert space: spin $\otimes$ boson, not spin alone. That tensor product is the step these notes deliberately do not take.

Model	Coupling term	Its single-spin ancestor
Jaynes–Cummings, Eq. (30)	$\hbar g(\hat{a}\hat{\sigma}_+ + \hat{a}^\dagger\hat{\sigma}_-)$	Section 4, field as a classical number
Holstein, Eq. (40)	$\hbar\omega g(\hat{a} + \hat{a}^\dagger)\hat{\sigma}_z$	Section 5, q as a classical parameter

In both cases the two-level machinery developed here survives intact — it is simply tensored with a harmonic oscillator. Keep the results boxed in [Sections 1](#) and [2](#); you will need every one of them again.

9 Problems

These are the problems that close the notebook. Numerical answers quoted in the solutions have been checked against direct diagonalization or propagation.

Problem 1. The excitonic dimer

Two chromophores sit close enough that their transition dipoles interact. In the singly-excited basis $\{|e_1g_2\rangle, |g_1e_2\rangle\}$ the Hamiltonian is

$$\hat{H} = \begin{bmatrix} \varepsilon_1 & J \\ J & \varepsilon_2 \end{bmatrix},$$

with J the excitonic coupling.

- Identify ε_0 , d_z and d_x .
- For identical chromophores, find the eigenstates and energies. The splitting $2|J|$ is the *Davydov splitting*.
- The transition dipole to the symmetric state is $\mu_1 + \mu_2$ and to the antisymmetric state $\mu_1 - \mu_2$. In the point-dipole approximation, $J > 0$ for a side-by-side (H-aggregate) geometry and $J < 0$ for head-to-tail (J-aggregate). In each case, is the *bright* state the upper or lower one, and does the absorption red- or blue-shift relative to the monomer?
- Adapt the code from Part 3 of the notebook to plot the two exciton energies and their oscillator strengths against $\varepsilon_1 - \varepsilon_2$ at fixed J .

Solution

(a) By inspection, $\varepsilon_0 = \frac{1}{2}(\varepsilon_1 + \varepsilon_2)$, $d_z = \varepsilon_1 - \varepsilon_2$, $d_x = 2J$, $d_y = 0$.

(b) Identical chromophores means $d_z = 0$, hence $\theta = \pi/2$ and the eigenstates are $|\pm\rangle = \frac{1}{\sqrt{2}}(|e_1g_2\rangle \pm |g_1e_2\rangle)$ with $E_{\pm} = \varepsilon \pm J$. Note the sign: the symmetric combination is the *upper* state when $J > 0$ and the *lower* state when $J < 0$.

(c) The oscillator strength goes as $|\mu_1 \pm \mu_2|^2$. For two parallel dipoles, the symmetric combination carries all of it and the antisymmetric combination is dark. The point-dipole coupling is $J = r^{-3}[\mu_1 \cdot \mu_2 - 3(\mu_1 \cdot \hat{r})(\mu_2 \cdot \hat{r})]$: side by side ($\mu \perp \hat{r}$) gives $J = +\mu^2/r^3 > 0$, head to tail ($\mu \parallel \hat{r}$) gives $J = -2\mu^2/r^3 < 0$.

Geometry	Sign of J	Bright (symmetric) state	Shift
H-aggregate (side by side)	$J > 0$	upper, at $\varepsilon + J$	blue
J-aggregate (head to tail)	$J < 0$	lower, at $\varepsilon + J$	red

This is why J-aggregates show the narrow, red-shifted, intense band that makes them useful as photographic sensitizers and light-harvesting mimics, while H-aggregates are

blue-shifted and comparatively dark.

(d) With $d_z \neq 0$ the two chromophores are inequivalent, the states are no longer exactly symmetric/antisymmetric, and the dark state steals intensity: the oscillator strengths become $|\cos(\theta/2)\mu_1 \pm \sin(\theta/2)\mu_2|^2$ and neither state is completely dark. Energetic disorder therefore *brightens* H-aggregates — a real and much-studied effect in molecular crystals.

Problem 2. Hückel butadiene is two dimers

The π system of *trans*-butadiene is a 4×4 Hückel problem ($H_{ii} = \alpha$, $H_{i,i+1} = \beta$), but it has a mirror symmetry exchanging $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$.

- Form $s_{1,2}$ and $a_{1,2}$, the symmetric and antisymmetric combinations $\frac{1}{\sqrt{2}}(p_1 \pm p_4)$ and $\frac{1}{\sqrt{2}}(p_2 \pm p_3)$, and show that $\hat{H}$ block-diagonalizes into two 2×2 blocks.
- Identify d_z and d_x for each block and use Eq. (7) to obtain all four π energies. Check against the textbook result $E = \alpha \pm 1.618\beta$, $\alpha \pm 0.618\beta$.
- Verify by diagonalizing the full 4×4 matrix numerically.

Solution

(a) Sites 1 and 4 are not bonded, so $H_{14} = 0$. Then

$$\langle s_1 | \hat{H} | s_1 \rangle = \frac{1}{2}(H_{11} + 2H_{14} + H_{44}) = \alpha, \quad \langle s_2 | \hat{H} | s_2 \rangle = \frac{1}{2}(H_{22} + 2H_{23} + H_{33}) = \alpha + \beta,$$

$$\langle s_1 | \hat{H} | s_2 \rangle = \frac{1}{2}(H_{12} + H_{13} + H_{42} + H_{43}) = \beta,$$

and every cross term $\langle s_i | \hat{H} | a_j \rangle$ vanishes because $\hat{H}$ commutes with the mirror operation while s and a belong to different eigenvalues of it. Repeating for the antisymmetric pair (the signs of H_{23} and H_{14} flip) gives

$$\hat{H}_S = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha + \beta \end{bmatrix}, \quad \hat{H}_A = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha - \beta \end{bmatrix}.$$

(b) Reading off the Pauli components:

$$\text{block } S: \quad \varepsilon_0 = \alpha + \frac{\beta}{2}, \quad d_z = -\beta, \quad d_x = 2\beta;$$

$$\text{block } A: \quad \varepsilon_0 = \alpha - \frac{\beta}{2}, \quad d_z = +\beta, \quad d_x = 2\beta.$$

Both have $|\mathbf{d}| = \sqrt{\beta^2 + 4\beta^2} = \sqrt{5}|\beta|$, so

$$E = \alpha \pm \frac{\beta}{2} \pm \frac{\sqrt{5}}{2}|\beta| = \alpha + 1.618\beta, \quad \alpha + 0.618\beta, \quad \alpha - 0.618\beta, \quad \alpha - 1.618\beta,$$

since $\frac{1}{2}(\sqrt{5} \pm 1) = 1.618, 0.618$ — the golden ratio and its reciprocal, which is where those famous Hückel numbers come from. With $\beta < 0$ the two lowest (occupied) levels are $\alpha + 1.618\beta$ (from block S) and $\alpha + 0.618\beta$ (from block A).

(c) Numerically, with $\alpha = 0$ and $\beta = -1$: the full matrix gives $\{-1.61803, -0.61803, +0.61803, +1.61803\}$, and the two blocks give $\{-1.61803, +0.61803\}$ and $\{-0.61803, +1.61803\}$ respectively — the same set.

Why this matters. Reducing a large Hamiltonian to independent small blocks using symmetry is one of the most useful manoeuvres in quantum chemistry, and here it turns a 4×4 problem into two problems you can solve with Eq. (7) in your head. The same trick underlies symmetry-adapted linear combinations throughout group theory.

Problem 3. Stückelberg oscillations

Modify `landau_zener_numeric` so that instead of one linear sweep, $d_z(t) = A \sin(\omega_{\text{sw}} t)$ oscillates back and forth through the crossing.

- Plot the population in $|\alpha\rangle$ against time over a few periods.
- Explain why it does not simply relax toward $1/2$.
- Find an amplitude and frequency for which the transfer after *two* passages is smaller than after one. What is interfering with what?

Solution

(a),(b) Each passage through the crossing acts as a beamsplitter, putting the system into a superposition of the two adiabatic surfaces with amplitudes fixed by Eq. (37). Between passages the two components accumulate *different* dynamical phases, $\Delta\phi = \hbar^{-1} \oint (E_+ - E_-) dt$, because the surfaces are separated. At the next passage the two amplitudes recombine and interfere. The dynamics is therefore a repeated interferometer, not a repeated random draw — so the population does not diffuse to $1/2$, it oscillates. These are Stückelberg oscillations, and the full sequence is a Landau–Zener–Stückelberg interferometer.

(c) Take $V_{DA} = 0.35$, $A = 6$ and scan ω_{sw} with $\hbar = 1$. The ratio (transfer after two passages)/(transfer after one) is strongly non-monotonic — it runs from about 10 down to about 1 and back as ω_{sw} varies over a range of order unity. Two representative points:

ω_{sw}	transfer after 1 passage	after 2 passages	ratio
0.450	0.041	0.430	10.4 (constructive)
0.700	0.0244	0.0241	0.99 (destructive)

At $\omega_{\text{sw}} \approx 0.700$ the second passage almost exactly undoes the first: the amplitude transferred on the way out and the amplitude transferred on the way back arrive π out of phase and cancel. Nothing has been “untransferred” in a classical sense — the two *amplitudes* interfere destructively. Changing ω_{sw} by a few percent changes $\Delta\phi$ by order π and flips the result, which is why the ratio swings so violently.

Problem 4. A genuine $\hat{\sigma}_y$: Peierls phases

A magnetic field attaches a phase to each hopping integral, $\beta \rightarrow \beta e^{i\phi_B}$.

- Write $\hat{H}$ for a two-site dimer with this hopping and identify d_x and d_y .
- Show that the eigenvalues are independent of ϕ_B , consistent with [Section 7](#).
- Now take a triangle of three sites, each bond carrying phase ϕ_B . Show that the eigenvalues *do* depend on the total flux $\Phi = 3\phi_B$ and that the dependence cannot be rephased away. Diagonalize numerically and plot the three levels against flux.

Solution

(a) With $H_{AB} = \beta e^{i\phi_B}$ and β real, $d_x = 2\beta \cos \phi_B$ and $d_y = -2\beta \sin \phi_B$, so $d_\perp = 2|\beta|$ regardless of ϕ_B : the phase only rotates $\mathbf{d}$ within the equatorial plane.

(b) By [Eq. \(7\)](#) the eigenvalues depend on the transverse components only through $d_\perp$, so $E_\pm = \varepsilon_0 \pm |\beta|$ for every ϕ_B . Numerically, $\phi_B = 0, 0.4, 1.1, 2.0$ all give $\{-|\beta|, +|\beta|\}$ exactly. Equivalently: one rephasing, $|B\rangle \rightarrow e^{i\phi_B} |B\rangle$, removes the phase completely.

(c) For the triangle with $t = \beta e^{i\phi_B}$ on each bond taken in a consistent circulation,

$$\hat{H} = \begin{bmatrix} 0 & t & t^* \\ t^* & 0 & t \\ t & t^* & 0 \end{bmatrix}, \quad E_n = 2\beta \cos\left(\frac{2\pi n + \Phi}{3}\right), \quad n = 0, 1, 2, \quad \Phi = 3\phi_B.$$

This is verified numerically: at $\Phi = 0$ the spectrum is $\{-2|\beta|, +|\beta|, +|\beta|\}$ with the familiar doubly degenerate level of a Hückel triangle; at $\Phi = \pi/2$ it is $\{-1.732, 0, +1.732\}|\beta|$; at $\Phi = \pi$ it is $\{-|\beta|, -|\beta|, +2|\beta|\}$. The degeneracy is lifted and the whole spectrum shifts with flux.

The counting argument is the point. Three sites admit three phase choices, but the overall phase is unobservable, leaving *two* useful rephasings against *three* bond phases. The combination that survives is the sum around the loop, $\Phi = \phi_{12} + \phi_{23} + \phi_{31}$, which is gauge invariant and physically is the magnetic flux threading the ring. A dimer has two sites and one bond — one usable rephasing, one phase — so nothing survives. That is the whole difference between (b) and (c), and it is the Aharonov–Bohm effect in its smallest possible setting.

Problem 5. Ramsey interferometry

Using the machinery of [Section 4](#), simulate the following on a system with detuning Δ : a $\pi/2$ pulse with phase $\varphi = 0$; free evolution with the drive off (so $\mathbf{d} = \hbar\Delta\hat{z}$) for a time τ ; a second $\pi/2$ pulse, also with $\varphi = 0$.

- Plot the final P_β against τ . What is the oscillation frequency?
- Repeat with the second pulse phase-shifted to $\varphi = \pi$. What happens, and why does this demonstrate that the *relative* phase of two pulses is physical even though the phase of one is not?
- This is the Ramsey separated-oscillatory-fields method, and it is how atomic clocks work. Why does a longer τ give a more precise frequency measurement?

Solution

Work in the hard-pulse limit $\omega_1 \gg \Delta$, so the pulses are effectively instantaneous rotations and the detuning acts only during τ .

(a) The first x -pulse takes $\mathbf{s}$ from the north pole to $-\hat{y}$ (Section 4). Free evolution rotates it in the equatorial plane at rate Δ , so $\mathbf{s}(\tau) = (\sin \Delta\tau, -\cos \Delta\tau, 0)$. The second x -pulse rotates about $\hat{x}$ by 90° in the same sense ($\hat{y} \rightarrow \hat{z}$), leaving $\mathbf{s} = (\sin \Delta\tau, 0, -\cos \Delta\tau)$. Hence

$$P_\beta(\tau) = \frac{1 - s_z}{2} = \frac{1 + \cos \Delta\tau}{2},$$

full-contrast fringes at the detuning frequency Δ . (Numerically, with $\omega_1 = 400\Delta$, the simulation matches this to better than 0.003.)

(b) With $\varphi = \pi$ the second rotation is about $-\hat{x}$, so $\hat{y} \rightarrow -\hat{z}$ and the final s_z changes sign: $P_\beta = \frac{1}{2}(1 - \cos \Delta\tau)$. The fringes are exactly inverted. Neither pulse phase is meaningful alone — calling the first one “ x ” was a convention — but the *difference* of the two phases determines whether you measure a fringe maximum or a minimum at $\tau = 0$. That is a measurable, convention-independent statement, and it is the concrete content of Section 7.

(c) The fringe pattern $\cos \Delta\tau$ has period $2\pi/\tau$ in Δ , so a frequency error $\delta\Delta$ produces a phase error $\delta\Delta \cdot \tau$. Doubling τ halves the detuning uncertainty for the same phase resolution: the precision improves as $1/\tau$. In practice τ is limited by decoherence, which is why buying a better clock means buying a longer coherence time.

Problem 6. The ammonia dipole paradox

At zero field the ground state of NH_3 is $\frac{1}{\sqrt{2}}(|L\rangle + |R\rangle)$, which has *zero* expectation value of the dipole operator — yet NH_3 is tabulated with $\mu = 1.47\text{ D}$.

- Compute $\langle \hat{\mu} \rangle$ in each of the two zero-field eigenstates, taking $\hat{\mu} = \mu \hat{\sigma}_z$ in the $\{|L\rangle, |R\rangle\}$ basis.
- A molecule prepared in $|L\rangle$ is not stationary. Using Eq. (14), find $\langle \hat{\mu} \rangle(t)$ and its oscillation frequency.
- Reconcile (a) and (b) with the tabulated value. On what timescale must a measurement act to see 1.47 D rather than zero?
- Deuterating to ND_3 drops the inversion splitting to about 0.053/cm. What happens to the crossover field $\mathcal{E}_{\text{cross}} = \Delta_t/\mu$, and why is the splitting so sensitive to isotopic substitution?

Solution

(a) With $\hat{\mu} = \mu \hat{\sigma}_z$ and eigenstates $|\pm\rangle = \frac{1}{\sqrt{2}}(|L\rangle \pm |R\rangle)$,

$$\langle \pm | \hat{\mu} | \pm \rangle = \frac{\mu}{2} (\langle L | \hat{\sigma}_z | L \rangle + \langle R | \hat{\sigma}_z | R \rangle) = \frac{\mu}{2}(1 - 1) = 0.$$

Both stationary states are exactly nonpolar. This is not an accident: they are parity eigenstates, and $\hat{\mu}$ is parity-odd, so its diagonal elements must vanish. The dipole operator

is purely *off*-diagonal in the energy eigenbasis — which is precisely why the 23.786 GHz transition between them is dipole allowed and drives the maser.

(b) $|L\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ has Bloch vector $\mathbf{s}(0) = +\hat{z}$, while $\mathbf{d} = -2\Delta_t\hat{x}$. By Eq. (14) the vector precesses in the zy plane at $|\mathbf{d}|/\hbar = 2\Delta_t/\hbar$, so

$$\langle\hat{\mu}\rangle(t) = \mu s_z(t) = \mu \cos\left(\frac{2\Delta_t}{\hbar}t\right),$$

oscillating at exactly the inversion frequency, 23.786 GHz (period 42 ps).

(c) Both are right; they answer different questions. A measurement slow compared with 42 ps averages Eq. (14) over many inversions and returns zero. A measurement fast compared with it — or any interaction that *localizes* the molecule before it can invert — sees the instantaneous 1.47 D. In practice a bulk dielectric measurement reports $\mu = 1.47$ D because collisions in the gas, and the local fields of neighbouring molecules in a liquid, both act far faster than 42 ps and keep the molecule pinned on one side. Figure 8 is the quantitative version of that statement: a field of order $\mathcal{E}_{\text{cross}}$ localizes the umbrella, and once localized the molecule has its full classical dipole. The “paradox” is really the statement that the tabulated dipole belongs to a *localized* molecule, not to an energy eigenstate of an isolated one.

(d) $\mathcal{E}_{\text{cross}}$ is proportional to Δ_t , so it falls by the same factor of roughly 15: from 16.07 kV/cm to about 1.1 kV/cm. ND₃ is far easier to localize with a field.

The sensitivity comes from the exponential dependence of a tunneling amplitude on the mass. In the WKB approximation $\Delta_t \sim \exp\left(-\hbar^{-1} \int \sqrt{2m[V(x) - E]} dx\right)$, so Δ_t falls roughly exponentially in $\sqrt{m}$. Deuterium is twice the mass of hydrogen, and although the relevant inversion coordinate involves the nitrogen moving against three hydrogens rather than a single H atom, the effective mass still increases enough to suppress the amplitude by more than an order of magnitude. This exponential mass sensitivity is the general signature of tunneling, and it is why kinetic isotope effects far larger than the classical $\sqrt{2}$ are taken as evidence that a reaction proceeds by tunneling.

A The Pauli algebra

Everything in Sections 1 and 2 rests on one product rule. Multiplying out the matrices directly,

$$\hat{\sigma}_x\hat{\sigma}_y = i\hat{\sigma}_z, \quad \hat{\sigma}_y\hat{\sigma}_z = i\hat{\sigma}_x, \quad \hat{\sigma}_z\hat{\sigma}_x = i\hat{\sigma}_y, \quad \hat{\sigma}_x^2 = \hat{\sigma}_y^2 = \hat{\sigma}_z^2 = \hat{\mathbb{1}}, \quad (48)$$

and each product in the first list reverses sign when the factors are exchanged. All of this is summarized by

Pauli product rule

$$\hat{\sigma}_i\hat{\sigma}_j = \delta_{ij}\hat{\mathbb{1}} + i\epsilon_{ijk}\hat{\sigma}_k, \quad (49)$$

with ϵ_{ijk} the Levi-Civita symbol and a sum over k implied.

Taking the symmetric and antisymmetric parts of Eq. (49) gives the commutator and anticommutator separately,

$$[\hat{\sigma}_i, \hat{\sigma}_j] = 2i \epsilon_{ijk} \hat{\sigma}_k, \quad \{\hat{\sigma}_i, \hat{\sigma}_j\} = 2\delta_{ij} \hat{\mathbb{1}}. \quad (50)$$

The first of these, written for $\hat{S}_x = \frac{\hbar}{2} \hat{\sigma}_x$ and friends, is the angular momentum algebra $[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z$ you verified numerically in Computational Set 1. The trace identities (4) follow by taking the trace of Eq. (49) and using $\text{Tr } \hat{\sigma}_k = 0$.

Proof of $(\mathbf{d} \cdot \hat{\sigma})^2 = |\mathbf{d}|^2 \hat{\mathbb{1}}$

Expand and apply Eq. (49):

$$\begin{aligned} (\mathbf{d} \cdot \hat{\sigma})^2 &= d_i d_j \hat{\sigma}_i \hat{\sigma}_j = d_i d_j (\delta_{ij} \hat{\mathbb{1}} + i \epsilon_{ijk} \hat{\sigma}_k) \\ &= |\mathbf{d}|^2 \hat{\mathbb{1}} + i \epsilon_{ijk} d_i d_j \hat{\sigma}_k = |\mathbf{d}|^2 \hat{\mathbb{1}}. \end{aligned} \quad (51)$$

The second term vanishes because ϵ_{ijk} is antisymmetric in $i \leftrightarrow j$ while $d_i d_j$ is symmetric, and the contraction of a symmetric with an antisymmetric tensor is zero. This is Eq. (6), and with it the eigenvalues (7) follow immediately: if $\hat{M}^2 = |\mathbf{d}|^2 \hat{\mathbb{1}}$ then every eigenvalue m of $\hat{M}$ satisfies $m^2 = |\mathbf{d}|^2$.

B The eigenvectors

Take $\varepsilon_0 = 0$ without loss of generality and write $\hat{H}$ in terms of the polar angles of Eq. (10):

$$\hat{H} = \frac{|\mathbf{d}|}{2} \begin{bmatrix} \cos \theta & \sin \theta e^{-i\varphi} \\ \sin \theta e^{+i\varphi} & -\cos \theta \end{bmatrix}. \quad (52)$$

Try $|+\rangle = (\cos \frac{\theta}{2}, e^{i\varphi} \sin \frac{\theta}{2})^T$. The first component of $\hat{H} |+\rangle$ is

$$\begin{aligned} \frac{|\mathbf{d}|}{2} [\cos \theta \cos \frac{\theta}{2} + \sin \theta e^{-i\varphi} e^{i\varphi} \sin \frac{\theta}{2}] &= \frac{|\mathbf{d}|}{2} [\cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2}] \\ &= \frac{|\mathbf{d}|}{2} \cos(\theta - \frac{\theta}{2}) = \frac{|\mathbf{d}|}{2} \cos \frac{\theta}{2}, \end{aligned} \quad (53)$$

using the cosine difference formula. The second component is, similarly,

$$\frac{|\mathbf{d}|}{2} e^{i\varphi} [\sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2}] = \frac{|\mathbf{d}|}{2} e^{i\varphi} \sin \frac{\theta}{2}. \quad (54)$$

So $\hat{H} |+\rangle = \frac{|\mathbf{d}|}{2} |+\rangle$, confirming Eq. (12) and $E_+ = \varepsilon_0 + |\mathbf{d}|/2$. The state $|-\rangle$ is obtained by $\theta \rightarrow \theta + \pi$, $\varphi \rightarrow \varphi + \pi$ (that is, $\mathbf{d} \rightarrow -\mathbf{d}$), which is why it appears with the opposite sign pattern.

Two checks worth doing yourself: $\langle + | - \rangle = 0$, and $\langle + | \hat{\sigma}_z | + \rangle = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta$, which together with the analogous x and y expectation values confirms $\mathbf{s}_+ = \hat{\mathbf{d}}$ as asserted in Section 2.3.

C The Bloch equation

Start from the Heisenberg equation for the expectation value of $\hat{\sigma}_a$,

$$\frac{d \langle \hat{\sigma}_a \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{\sigma}_a] \rangle. \quad (55)$$

The identity part of $\hat{H}$ commutes with everything, so only $\frac{1}{2} d_b \hat{\sigma}_b$ contributes:

$$[\hat{H}, \hat{\sigma}_a] = \frac{1}{2} d_b [\hat{\sigma}_b, \hat{\sigma}_a] = \frac{1}{2} d_b (2i \epsilon_{bac} \hat{\sigma}_c) = i d_b \epsilon_{bac} \hat{\sigma}_c, \quad (56)$$

using Eq. (50). Therefore

$$\frac{d \langle \hat{\sigma}_a \rangle}{dt} = \frac{i}{\hbar} \cdot i d_b \epsilon_{bac} \langle \hat{\sigma}_c \rangle = -\frac{1}{\hbar} \epsilon_{bac} d_b \langle \hat{\sigma}_c \rangle = +\frac{1}{\hbar} \epsilon_{abc} d_b \langle \hat{\sigma}_c \rangle, \quad (57)$$

where the last step used $\epsilon_{bac} = -\epsilon_{abc}$. The right-hand side is precisely the a -component of a cross product, $(\mathbf{d} \times \mathbf{s})_a = \epsilon_{abc} d_b s_c$, giving Eq. (14):

$$\frac{d\mathbf{s}}{dt} = \frac{1}{\hbar} \mathbf{d} \times \mathbf{s}. \quad (58)$$

Remark. For a real spin in a real magnetic field, $\mathbf{d} = -\gamma \hbar \mathbf{B}$ and this becomes $\dot{\mathbf{s}} = -\gamma \mathbf{B} \times \mathbf{s}$ — the classical equation for a magnetic moment precessing in a field, and the reason the same picture works for a classical gyroscope. That a fully quantum two-level system obeys a *closed, classical-looking* equation of motion is special to two dimensions: the Bloch vector contains all the information in the state, so no higher moments are needed to close the hierarchy.

D The rotating-frame transformation

Let $R(t) = e^{+i\omega t \hat{\sigma}_z/2}$ and $|\tilde{\psi}\rangle = R|\psi\rangle$. Then

$$\begin{aligned} i\hbar \frac{d|\tilde{\psi}\rangle}{dt} &= i\hbar \dot{R}|\psi\rangle + R \left(i\hbar \frac{d|\psi\rangle}{dt} \right) = i\hbar \left(\frac{i\omega}{2} \hat{\sigma}_z \right) R|\psi\rangle + R \hat{H}|\psi\rangle \\ &= \left[R \hat{H} R^\dagger - \frac{\hbar\omega}{2} \hat{\sigma}_z \right] |\tilde{\psi}\rangle \equiv \hat{H}_{\text{rot}} |\tilde{\psi}\rangle. \end{aligned} \quad (59)$$

So we need $R \hat{H} R^\dagger$. Since R is built from $\hat{\sigma}_z$, the $\hat{\sigma}_z$ term passes through untouched. For the transverse operators, define $\hat{A}(\vartheta) = e^{i\vartheta \hat{\sigma}_z/2} \hat{\sigma}_x e^{-i\vartheta \hat{\sigma}_z/2}$ and differentiate:

$$\frac{d\hat{A}}{d\vartheta} = \frac{i}{2} [\hat{\sigma}_z, \hat{A}], \quad \left. \frac{d\hat{A}}{d\vartheta} \right|_0 = \frac{i}{2} (2i\hat{\sigma}_y) = -\hat{\sigma}_y. \quad (60)$$

The analogous calculation for $\hat{B}(\vartheta)$ built from $\hat{\sigma}_y$ gives $\hat{B}'(0) = +\hat{\sigma}_x$, so the pair satisfies $\hat{A}' = -\hat{B}$, $\hat{B}' = +\hat{A}$ and

$$R \hat{\sigma}_x R^\dagger = \hat{\sigma}_x \cos \omega t - \hat{\sigma}_y \sin \omega t, \quad R \hat{\sigma}_y R^\dagger = \hat{\sigma}_y \cos \omega t + \hat{\sigma}_x \sin \omega t. \quad (61)$$

Now substitute Eq. (61) into the drive term of Eq. (22). Write $c = \cos(\omega t + \varphi)$, $s = \sin(\omega t + \varphi)$, $C = \cos \omega t$, $S = \sin \omega t$:

$$\begin{aligned} \frac{\hbar\omega_1}{2} \left[c (\hat{\sigma}_x C - \hat{\sigma}_y S) + s (\hat{\sigma}_y C + \hat{\sigma}_x S) \right] &= \frac{\hbar\omega_1}{2} \left[\hat{\sigma}_x (cC + sS) + \hat{\sigma}_y (sC - cS) \right] \\ &= \frac{\hbar\omega_1}{2} \left[\hat{\sigma}_x \cos \varphi + \hat{\sigma}_y \sin \varphi \right], \end{aligned} \quad (62)$$

where the two brackets collapsed by the cosine and sine difference formulas evaluated at $(\omega t + \varphi) - \omega t = \varphi$. **All the time dependence has cancelled.** Combining with the $\hat{\sigma}_z$ terms from Eq. (59),

$$\hat{H}_{\text{rot}} = \frac{\hbar\omega_0}{2} \hat{\sigma}_z - \frac{\hbar\omega}{2} \hat{\sigma}_z + \frac{\hbar\omega_1}{2} (\cos \varphi \hat{\sigma}_x + \sin \varphi \hat{\sigma}_y), \quad (63)$$

which is Eq. (23).

What was given up. Nothing was approximated — for a genuinely *circularly* rotating drive this transformation is exact. (For a linearly polarized drive, $\propto \cos \omega t \hat{\sigma}_x$, one first decomposes into two counter-rotating components and discards the far-off-resonant one; that discard is the rotating-wave approximation, and it is a genuine approximation.) What is lost is directness of interpretation: $|\tilde{\psi}\rangle$ is not the physical state, and a stationary Bloch vector in the rotating frame corresponds to one spinning at ω in the laboratory.

E The Marcus barrier

From Eq. (31) with $\lambda = 2kq_0^2$, the diabats cross where $V_D(q^\ddagger) = V_A(q^\ddagger)$:

$$\frac{1}{2}k(q + q_0)^2 = \frac{1}{2}k(q - q_0)^2 + \Delta G^\circ \implies 2kq_0q = \Delta G^\circ \implies q^\ddagger = \frac{\Delta G^\circ}{2kq_0} = \frac{\Delta G^\circ q_0}{\lambda}. \quad (64)$$

The activation energy is the diabatic energy there, measured from the donor minimum at $q = -q_0$ where $V_D = 0$:

$$\begin{aligned} \Delta G^\ddagger = V_D(q^\ddagger) &= \frac{1}{2}k \left(\frac{\Delta G^\circ q_0}{\lambda} + q_0 \right)^2 = \frac{1}{2}kq_0^2 \left(\frac{\Delta G^\circ + \lambda}{\lambda} \right)^2 \\ &= \frac{\lambda}{4} \cdot \frac{(\lambda + \Delta G^\circ)^2}{\lambda^2} = \frac{(\lambda + \Delta G^\circ)^2}{4\lambda}, \end{aligned} \quad (65)$$

using $\frac{1}{2}kq_0^2 = \lambda/4$. This is Eq. (35). Three limits are worth memorizing:

$$\Delta G^\circ = 0 \implies \Delta G^\ddagger = \frac{\lambda}{4}, \quad \Delta G^\circ = -\lambda \implies \Delta G^\ddagger = 0, \quad \Delta G^\circ = -2\lambda \implies \Delta G^\ddagger = \frac{\lambda}{4}. \quad (66)$$

The barrier is symmetric about $\Delta G^\circ = -\lambda$: a reaction that is *too* exergonic by exactly λ is as slow as one with no driving force at all.

Note finally that this is the barrier on the *diabatic* surface. The true barrier on the lower adiabatic surface is lower by roughly V_{DA} near the crossing. When $V_{DA} \ll k_B T$ the nonadiabatic (golden-rule) rate applies and the diabatic barrier is the right one to use; when $V_{DA} \gg k_B T$ the reaction is adiabatic and the transition state is the top of $E_-(q)$. The dimensionless ratio that decides which regime you are in is essentially the Landau-Zener exponent of Eq. (37).

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